Buy Now, Pay Later (BNPL) Risk Prediction Model

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1 Introduction

1.1 Background

Buy Now, Pay Later (BNPL) services have become increasingly popular as a flexible payment option in modern retail. These services allow customers to purchase products immediately and pay for them later in smaller installments, rather than paying the full amount upfront. This approach makes shopping more convenient and accessible, especially for individuals who may not prefer traditional credit options.

The growing use of BNPL is closely linked to the rise of online shopping and digital payment platforms. Many retailers now offer BNPL at checkout, making it an easy and attractive choice for consumers. As a result, more people are relying on this payment method for everyday purchases as well as larger expenses.

However, this convenience can also influence spending behavior. Users may feel that items are more affordable than they actually are because payments are split into smaller amounts. This can encourage impulse buying and increase the likelihood that individuals take on multiple payment plans at the same time.

1.2 Problem Description

Despite its benefits, BNPL introduces important financial risks for both users and providers. Some users may struggle to keep up with their installment payments, especially when managing several plans at once. This can lead to missed or delayed payments which may negatively affect their financial situation.

For companies offering BNPL services, it is difficult to predict which users are likely to miss payments. Without accurate predictions, lenders may take on too much risk or make decisions that are not well-informed. This creates challenges in balancing financial performance with responsible lending practices.

In addition, there is a need to ensure that any prediction system is fair and transparent. Decisions that rely on automated methods must avoid bias and should clearly explain why certain users are considered higher risk. Therefore, this project develops a reliable and understandable approach to predicting repayment behavior for BNPL.

1.3 Questions to be Addressed

This project aims to address the following questions:

1. How can BNPL repayment risk be modeled in a way that is clear and easy to interpret?
2. How can uncertainty in consumer credit risk predictions be measured and communicated?
3. Which factors are most strongly associated with delayed or missed payments?

4. How can ethical concerns be addressed when using automated systems for BNPL risk assessment?
5. How can a responsible and reproducible approach to financial modeling be developed using tools like Python and collaborative platforms?

2 Data

2.1 Description

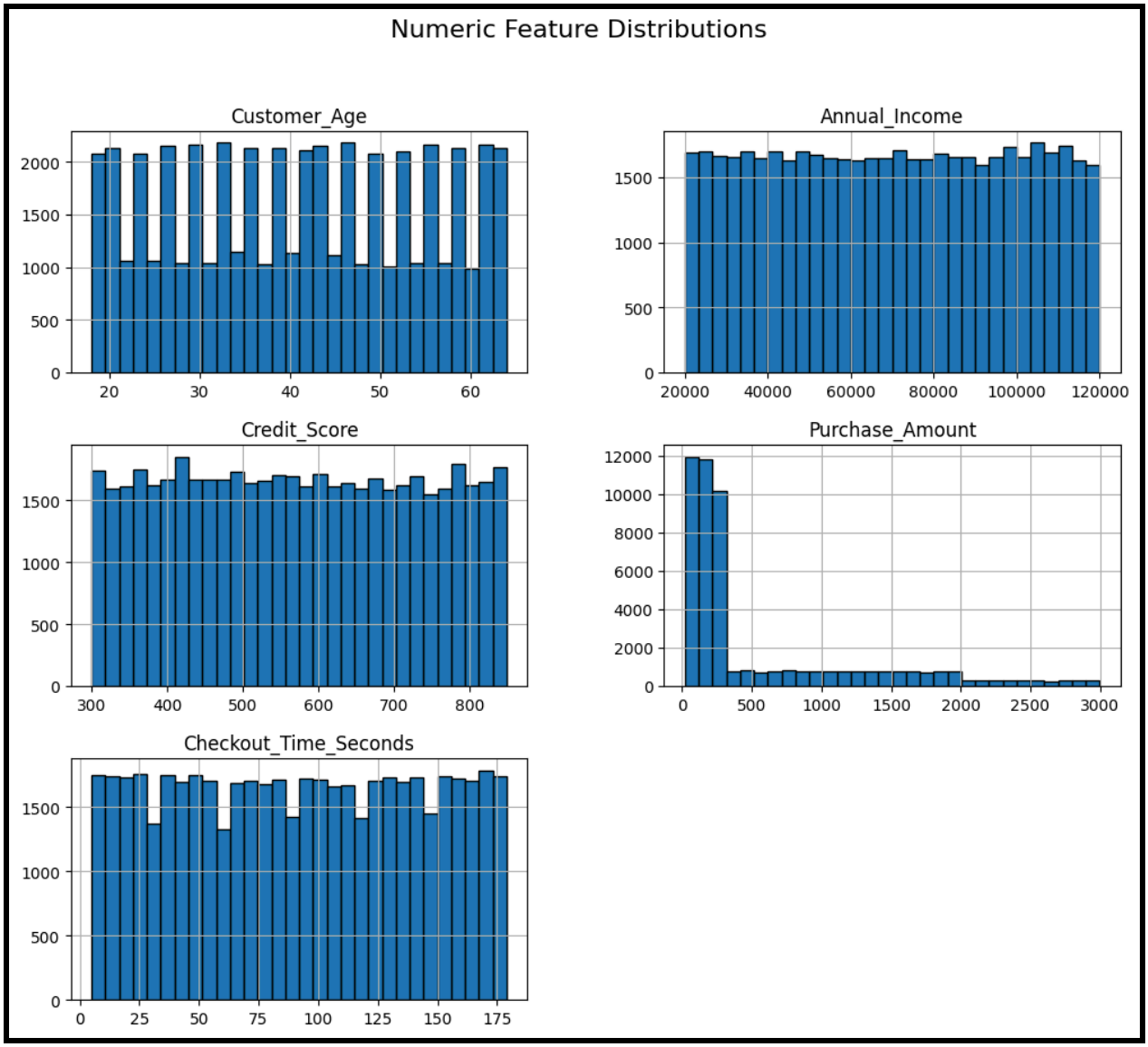
The data used in this project comes from a synthetic Buy Now, Pay Later (BNPL) dataset obtained from Kaggle. The raw dataset contains 50,000 observations, where each row represents a single customer transaction. A synthetic dataset was used because real BNPL transaction data is not publicly available. Companies such as Klarna and Afterpay do not release detailed customer-level repayment data due to privacy concerns and the potential impact on users' financial and credit information. By using synthetic data, this project is able to preserve user privacy and avoid ethical concerns related to handling sensitive financial records, while still working with data that reflects realistic consumer and repayment patterns. The dataset was generated in Python using the Faker library and was designed to mimic real-world BNPL behavior.

2.2 Variables (for Raw Dataset)

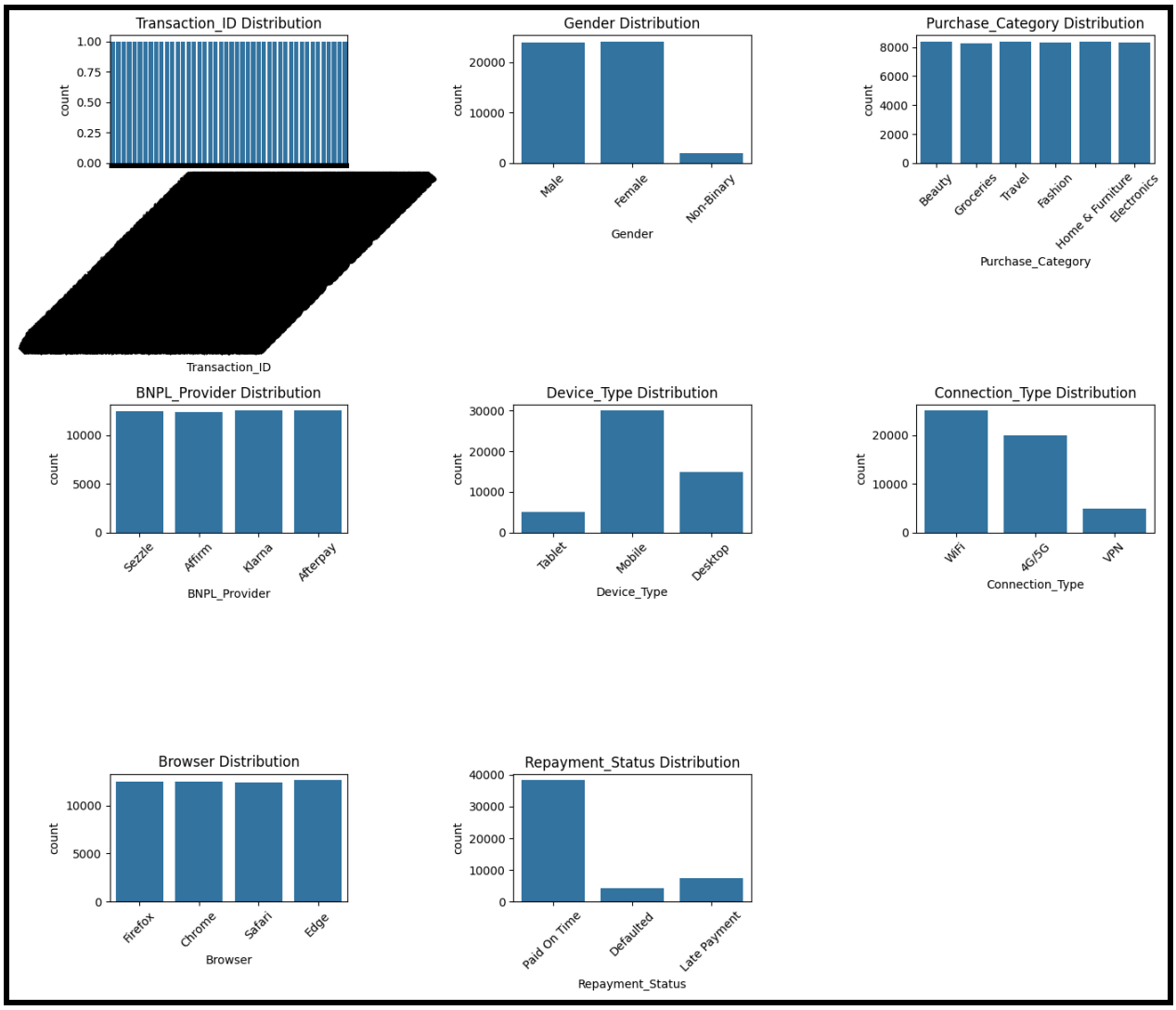
The response variable in the dataset is `Repayment_Status`, which records the repayment outcome for each transaction as Paid On Time, Late Payment, or Defaulted. The dataset includes five numerical predictors: `Annual_Income`, `Credit_Score`, `Purchase_Amount`, `Customer_Age`, and `Checkout_Time_Seconds`. It also includes seven categorical predictors: `Transaction_ID`, which is a unique identifier for each transaction, `Gender`, `Purchase_Category`, `BNPL_Provider`, `Device_Type`, `Connection_Type`, and `Browser`. The categories represented in these variables include values such as Female, Male, and Non-Binary for gender; Electronics, Fashion, Groceries, Home & Furniture, and Travel for purchase category; Affirm, Afterpay, Klarna, and Sezzle for BNPL provider; Desktop, Mobile, and Tablet for device type; 4G/5G, VPN, and WiFi for connection type; and Chrome, Edge, Firefox, and Safari for browser. Together, these variables provide information about customer financial characteristics, transaction details, and digital behavior that may be related to repayment risk.

```
First 5 rows:
                         Transaction_ID  Customer_Age Gender  Annual_Income  \
0  705eef54-52b2-4912-b562-d9f7b4184a6d            56   Male          32293
1  7dbe58a0-0177-4572-8f3a-566d907e9c56            46   Male          72774
2  b77e4be3-bf88-4bb0-addb-45d84e6a3f65            32   Male          82207
3  5c3c24a8-675e-46fd-8312-4ac37cbb97c5            60   Male          92498
4  4bb92076-f15e-4cfc-8aac-58395d69aedc            25   Male          32060

   Credit_Score Purchase_Category BNPL_Provider  Purchase_Amount Device_Type  \
0           353            Beauty        Sezzle              249      Tablet
1           354         Groceries        Affirm              188      Mobile
2           630            Travel        Sezzle             1610     Desktop
3           470           Fashion        Sezzle              120      Mobile
4           502            Travel        Klarna             1849      Mobile

  Connection_Type  Checkout_Time_Seconds  Browser Repayment_Status
0            WiFi                     82  Firefox     Paid On Time
1            WiFi                     60  Firefox     Paid On Time
2            WiFi                     86   Chrome     Paid On Time
3           4G/5G                    169   Chrome     Paid On Time
4           4G/5G                     38   Chrome        Defaulted
```

Sample of the raw dataset with all original variables (first 5 observations)



Distributions of the numeric features in the raw dataset



Distributions of the categorical features in the raw dataset

2.3 Preprocessing

Before fitting the model, several preprocessing steps were carried out to prepare the dataset for analysis. First, the variable `Transaction_ID` was removed because it serves only as a unique identifier and does not provide meaningful predictive information about repayment behavior. The dataset was then cleaned by checking for duplicate rows and missing values to ensure that the analysis was based on complete and reliable observations. Next, the response variable `Repayment_Status` was transformed from a three-category outcome into a binary outcome, where 0 represents transactions that were Paid On Time and 1 represents transactions with either a Late Payment or a Defaulted status. This transformation makes the outcome suitable for binary classification modeling.

The categorical predictors were converted into numerical form using index variable encoding so that they could be incorporated into the statistical model. In addition, the numerical predictors were standardized to have mean 0 and variance 1. This scaling step was important to place variables measured on different scales onto a common basis and to support regularization-based feature selection. After preprocessing, the data was split into training and testing sets, with 80% of the observations used for model fitting (training) and 20% reserved for out-of-sample evaluation (testing). To reduce model complexity and retain the most informative predictors, feature selection was performed using Lasso, which applies L1 regularization to shrink less important coefficients toward zero. The final training and testing datasets were then defined as X_{train}, y_{train}, X_{test}, y_{test}, using only the selected features.

3 Formulation of the mathematical model

3.1 Set-up of Variables and Parameters

The model that we are interested in is:

$$y_i \sim Bernoulli(p_i)$$
$$logit(p_i) = \alpha + X_i\beta$$
$$\alpha \sim Normal(0, 5)$$
$$\beta_j \sim Normal(0, 2)$$

where:

- y_i is the binary response variable, representing the repayment status for transaction i and equal to 0 if Paid On Time and 1 if Late/Defaulted
- p_i is the probability that transaction i was Late/Defaulted
- X_i is the vector of predictor variables for transaction i
- α represents the intercept parameter with a specified prior probability distribution that is Normal with a mean of 0 and standard deviation of 5
- β represents the vector of coefficients parameter with each β_j has a specified prior probability distributions that is Normal with a mean of 0 and standard deviation of 2

3.2 Assumptions

The proposed Bayesian logistic regression model relies on several standard assumptions. First, after preprocessing, the repayment outcome is binary, so each transaction is classified as either 0 for paid on time or 1 for late payment or default. Second, each row in the dataset is treated as an independent BNPL transaction, meaning the repayment outcome of one transaction is assumed not to directly affect the outcome of another. Third, the model assumes that the predictors influence repayment risk linearly on the log-odds scale. In other words, each predictor has an additive effect on the logarithm of the odds of repayment failure. The model also assumes that there is no perfect multicollinearity among the predictors, so no predictor can be written as an

exact linear combination of the others. Finally, the same coefficient vector β is assumed to apply across all transactions in the dataset, meaning that the relationship between the predictors and repayment risk is taken to be constant throughout the sample.

3.3 Restrictions

There are also some practical restrictions. Since the model is based on logistic regression, the response must remain binary. The predictors must be numerically represented, which is why categorical variables were encoded before fitting the model. In addition, the interpretation of the coefficients depends on the standardized and encoded form of the predictors rather than their original raw units. Another practical restriction is potential class imbalance in the response variable, since late or missed payments may occur much less frequently than on-time payments, which can make the model more sensitive to the majority class and reduce performance for identifying higher-risk cases.

target	count
0	38399
1	11601

dtype: int64

Imbalance distribution in dataset (default vs non default)

3.4 Justification

These modeling choices are appropriate for the problem and the processed data. The Bernoulli likelihood is suitable because the repayment status was re-coded as a binary variable taking values 0 (non-default) and 1 (default), which matches the Bernoulli distribution's two possible outcomes, 0 and 1. The logit link is also appropriate because the goal of the project is to model the probability of repayment failure while ensuring that all predicted probabilities remain between 0 and 1. For the priors, the intercept is assigned a $Normal(0, 5)$ prior, which is broad on the log-odds scale and therefore allows for a wide range of baseline repayment failure probabilities. The regression coefficients are assigned $Normal(0, 2)$ priors. Since the predictors were standardized during preprocessing, this prior places most of its probability mass on reasonable effect sizes on the log-odds scale. This choice is sensible because the predictors do not share a natural physical unit, so the priors are based on the standardized scale created during preprocessing rather than on original domain-specific units. Overall, this framework is consistent with the project goal of building an interpretable Bayesian model for BNPL repayment risk.

4 Solution of the problem

4.1 Problem Formulation and Modelling Strategy

The objective of this project is to predict the **probability of customer default** in a *Buy Now, Pay Later (BNPL)* system using historical transaction records and behavioral features.

To address this problem, we adopt a **two-stage modelling strategy**:

1. **Classical Logistic Regression:**
   A baseline model is developed using classical logistic regression, estimated via *maximum likelihood estimation (MLE)*. This approach provides an interpretable benchmark for evaluating predictive performance and understanding key factors influencing default likelihood.
2. **Bayesian Logistic Regression:**
   Next, we extend the analysis using a Bayesian framework, where model parameters are treated as random variables with prior distributions. Parameter inference is conducted using ***Hamiltonian Monte Carlo (HMC)*** implemented with the ***No-U-Turn Sampler (NUTS)*** to efficiently explore the posterior distribution.

This combined approach balances predictive performance and interpretability. While the classical logistic regression model offers a transparent, easy-to-communicate baseline, the Bayesian framework adds a layer of **probabilistic interpretability** - allowing for explicit **uncertainty quantification** in both parameter estimates and predictions. This capability is particularly valuable in **financial risk modelling**, where understanding the confidence and variability behind predictions is essential.

4.2 Mathematical Foundation of Logistic Regression

Logistic regression models the conditional probability of default using the logistic (sigmoid) link function:

$$P(Y = 1|X) = \frac{1}{1+e^{-z}}$$

where the linear predictor is defined as:

$$z = \beta_0 + X^T\beta$$

Here, β_0 denotes the intercept and β is the vector of regression coefficients associated with the input features.

An equivalent formulation is given in terms of the log-odds (logit transformation):

$$log\left(\frac{P(Y=1|X)}{1-P(Y=1|X)}\right) = \beta_0 + X^T\beta$$

In this representation, each coefficient β_j corresponds to the change in the log-odds of default for a one-unit increase in feature X_j, holding all other variables constant. This property provides strong interpretability, which is particularly valuable in financial risk modelling.

The model parameters β are estimated using Maximum Likelihood Estimation (MLE), where the likelihood is derived from the Bernoulli distribution of the binary response variable. Since no closed-form solution exists, the optimization is performed using numerical methods implemented in Python.

The trained model was evaluated using multiple performance metrics, including accuracy, ROC-AUC, precision, recall, and F1-score. In addition to the standard classification threshold of 0.5, alternative thresholds were explored to analyze the trade-off between precision and recall.

This is particularly important in the presence of class imbalance, where accurately identifying high-risk (default) customers is a priority.

Overall, logistic regression provides a strong and interpretable baseline model, capturing the relationship between features and default probability while serving as a benchmark for comparison with the Bayesian approach.

4.3 Parameter Estimation

Let $p_i = P\left(Y_i = 1 | X_i\right)$ denote the predicted probability of default for observation i. Under the assumption of conditional independence, the likelihood function is given by:

$$L(\beta) = \prod_{i=1}^{n} p_i^{y_i}\left(1 - p_i\right)^{1-y_i}$$

Taking logs yields the log-likelihood:

$$l(\beta) = \sum_{i=1}^{n}\left[y_i log p_i + \left(1 - y_i\right) log\left(1 - p_i\right)\right]$$

The parameter vector β is estimated via Maximum Likelihood Estimation (MLE) by numerically maximizing $l(\beta)$, typically using gradient-based optimization methods because no closed-form solution exists. The resulting model provides a deterministic estimate of default probabilities and serves as a strong, interpretable baseline for comparison.

4.4 Bayesian Logistic Regression Framework

Classical logistic regression produces point estimates for parameters and predictions but does not directly account for parameter uncertainty. To address this limitation, we adopt a Bayesian logistic regression framework, treating parameters as random variables with prior distributions.

```
# Bayesian Logistic Regression Model

with pm.Model() as bayes_logistic_model:

    intercept = pm.Normal("intercept", mu=0, sigma=5)

    coefficients = pm.Normal(
        "coefficients",
        mu=0,
        sigma=2,
        shape=X_train_scaled.shape[1]
    )

    logits = intercept + pm.math.dot(X_train_scaled, coefficients)

    likelihood = pm.Bernoulli(
        "likelihood",
        logit_p=logits,
        observed=y_train_array
    )
```

Code Snippet: Bayesian logistic regression model specification with Gaussian priors on coefficients.

4.5 Prior Specification

We assign Gaussian priors to the intercept and slope coefficients:

$$\beta_j \sim N\left(0, \sigma^2\right),\ \beta_0 \sim N\left(0, \sigma^2\right)$$

These priors encode weakly informative beliefs centered at zero and act as a regularization mechanism, shrinking extreme coefficients and helping to mitigate overfitting in high-dimensional feature spaces.

4.6 Likelihood

The likelihood for the Bayesian model is identical to the classical logistic regression likelihood. For each observation i:

$$Y_i \sim Bernoulli(p_i), p_i = \sigma\left(\beta_0 + \beta^T X)i\right)$$

This connects the binary outcome to the predictors via the logistic function.

4.7 Posterior Distribution

Using Bayes' theorem, the posterior distribution over parameters is:

$$P(\beta|X, Y) = \frac{P(Y|X,\beta)P(\beta)}{P(Y|X)}$$

Since the normalizing constant $P(Y|X)$ is intractable for logistic models, we work with the unnormalized posterior:

$$P(\beta|X, Y) \propto P(Y|X, \beta)P(\beta)$$

This posterior combines information from the data (via the likelihood) and prior beliefs (via the priors), yielding a full probabilistic characterization of parameter uncertainty

4.8 Posterior Inference via MCMC

The posterior distribution in Bayesian logistic regression does not admit a closed-form solution due to the non-conjugacy of the logistic likelihood with Gaussian priors. Therefore, we approximate the posterior using Markov Chain Monte Carlo (MCMC) methods.

4.9 Hamiltonian Monte Carlo (HMC)

Hamiltonian Monte Carlo (HMC) improves sampling efficiency by introducing auxiliary momentum variables and simulating Hamiltonian dynamics to propose new states. In this framework:

- The position corresponds to the parameter vector β
- The momentum corresponds to auxiliary variables introduced to define a Hamiltonian system.

By following approximate Hamiltonian trajectories, HMC is able to make long-distance moves in parameter space while maintaining a high acceptance probability, greatly reducing random-walk behavior compared to simpler MCMC schemes such as random-walk Metropolis.

4.10 No-U-Turn Sampler (NUTS)

We further employ the No-U-Turn Sampler (NUTS), an adaptive extension of HMC that automatically determines the number of leapfrog steps (trajectory length) and tunes step sizes during warm-up. NUTS:

- Eliminates the need for manual tuning of trajectory length.
- Adapts key hyperparameters to improve convergence and sampling efficiency.
- Avoids trajectories that double back ("U-turn"), which would waste computation and degrade mixing.

The MCMC procedure produces a sequence of draws:

$$\left\{\beta^{(1)}, \beta^{(2)}, \ldots\ldots, \beta^{(S)}\right\}$$

which approximate samples from the posterior distribution $P(\beta|X, Y)$.

```
# ============================================================
# Posterior sampling with NUTS (adaptive HMC)
# ============================================================

with bayes_logistic_model:
    trace = pm.sample(
        draws=1000,
        tune=1000,
        chains=2,
        cores=1,
        target_accept=0.9,
        random_seed=42,
        return_inferencedata=True
    )
```

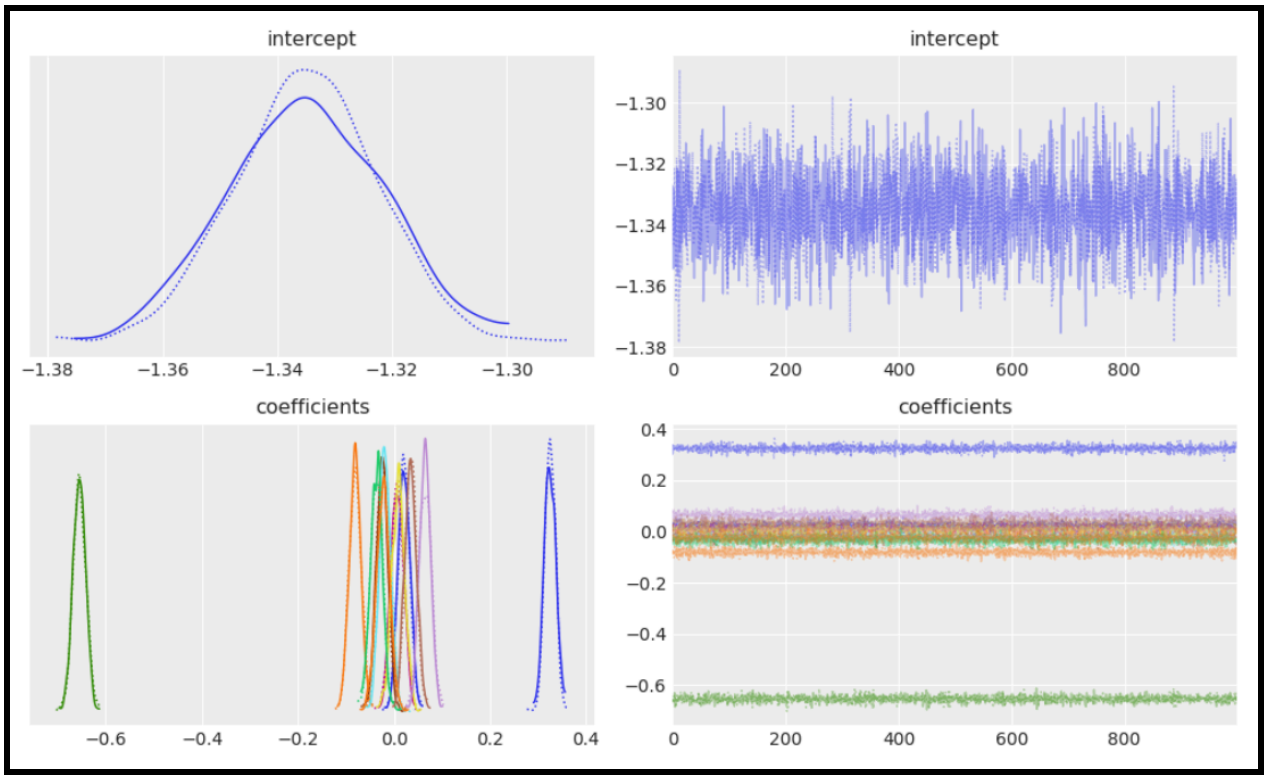
Progress	Draw	Divergences	Step size	Grad evals	Speed	Elapsed	Remaining
————————	2000	0	0.478	7	15.85 draws/s	0:02:06	0:00:00
————————	2000	0	0.606	7	8.74 draws/s	0:03:48	0:00:00

<u>Code + Output:</u> Posterior sampling performed using Hamiltonian Monte Carlo with the No-U-Turn Sampler (NUTS).

<u>4.11 Convergence Diagnostics and Model Stability</u>

To ensure the reliability of posterior inference, we conduct standard MCMC convergence diagnostics:

- Visual inspection of trace plots to assess mixing, stationarity, and absence of pathological behavior.
- Divergence checks to detect numerical instabilities and problematic posterior geometries.
- Posterior summaries (means, variances, and credible intervals) to confirm reasonable parameter behavior.



Trace plots for selected parameters showing good mixing, stationarity, and absence of pathological behavior across MCMC chains.

The trace plots demonstrate good mixing and convergence of the Markov chains for both the intercept and coefficient parameters. The chains fluctuate around stable mean values without exhibiting systematic trends or drifts, indicating stationarity. Additionally, the absence of large jumps or irregular patterns suggests that the sampler is efficiently exploring the posterior distribution. The density plots further show well-defined, unimodal posterior distributions, indicating stable and consistent parameter estimates. Together, these diagnostics confirm that the MCMC sampling procedure has converged and that the posterior estimates are reliable for inference and prediction.

A key outcome of the final model is that we obtain zero divergent transitions, indicating numerically stable computation, well-behaved posterior geometry under the chosen priors and parameterization, and reliable exploration of the posterior by HMC/NUTS. This is particularly important for HMC-based methods, where divergences often signal model misspecification or poor scaling.

4.12 Predictive Inference

Unlike classical logistic regression, which yields single point predictions for the default probability, the Bayesian model generates a distribution over predictions by integrating parameter uncertainty.

4.13 Posterior Mean Prediction

A simple way to obtain a point prediction from the Bayesian model is to use the posterior mean of the parameters:

$$\hat{p} = \sigma\left(E\left[\beta_0\right] + E[\beta]^T X\right)$$

This yields point estimates of default probabilities that are directly comparable to those from classical logistic regression while still being informed by the posterior distribution.

```
# Bayesian predictions using posterior mean coefficients
# With threshold = 0.5

posterior_intercept = trace.posterior["intercept"].mean(dim=("chain", "draw")).values.item()
posterior_coeffs = trace.posterior["coefficients"].mean(dim=("chain", "draw")).values

# posterior_coeffs
posterior_coeffs = np.ravel(posterior_coeffs)

logits_test = posterior_intercept + np.dot(X_test_scaled, posterior_coeffs)
probs_test = 1 / (1 + np.exp(-logits_test))
preds_test = (probs_test >= 0.5).astype(int)

print("Bayesian Logistic Regression Results")
print("-----------------------------------")
print("Accuracy :", round(accuracy_score(y_test_array, preds_test), 4))
print("ROC-AUC  :", round(roc_auc_score(y_test_array, probs_test), 4))

print("\nClassification Report:")
print(classification_report(y_test_array, preds_test))
```

Code Snippet: Posterior mean coefficients are used to compute predicted probabilities, followed by threshold-based classification for decision-making.

4.14 Posterior Predictive Distribution

A more complete Bayesian approach integrates over the entire posterior distribution of parameters:

$$P(Y = 1|X) = \int \sigma\left(\beta_0 + \beta^T X\right) P(\beta|X, Y) d\beta$$

We approximate this integral using the MCMC samples:

$$\hat{p} \approx \frac{1}{S} \sum_{s=1}^{S} \sigma\left(\beta_0^{(s)} + \beta^{(s)T} X\right)$$

This posterior predictive distribution captures uncertainty in both parameters and outcomes, which is crucial for risk-sensitive decisions such as setting credit limits, pricing BNPL products, and establishing risk thresholds.

```
logits_test = posterior_intercept + np.dot(X_test_scaled, posterior_coeffs)
probs_test = 1 / (1 + np.exp(-logits_test))
preds_test = (probs_test >= 0.3).astype(int)
```

Code Snippet: Posterior mean coefficients are used to compute predicted probabilities via the logistic function, which are then converted into class labels using a decision threshold of 0.3.

4.15 Feature Engineering and Scaling

The models are trained on a feature-engineered dataset constructed from raw BNPL transaction records and user behavior. Transactional and behavioral variables are transformed into meaningful predictors, such as aggregated spending patterns, repayment behavior, and temporal statistics, to better capture underlying credit risk profiles.

To facilitate efficient optimization and stable sampling, all continuous features are standardized:

$$X_{scaled} = \frac{X-\mu}{\sigma}$$

where μ and σ denote the empirical mean and standard deviation of each feature. Standardization improves numerical stability during gradient-based optimization in logistic regression. Enhances the geometry of the posterior for HMC/NUTS, leading to more efficient sampling. Ensures that features are on a comparable scale, which aids regularization and interpretability.

In conclusion, this modelling framework provides a comprehensive and theoretically grounded approach to BNPL default prediction by integrating both classical and Bayesian methodologies. The logistic regression model offers a strong, interpretable baseline, while the Bayesian extension enhances the model by incorporating parameter uncertainty and enabling probabilistic inference. Through the use of MCMC methods, specifically HMC with NUTS, the posterior distribution is efficiently approximated, ensuring reliable and stable estimation as confirmed by convergence diagnostics. This combination of interpretability, predictive capability, and uncertainty quantification makes the proposed approach well-suited for financial risk modelling, where both accuracy and confidence in predictions are critical for informed decision-making.

5 Interpretation of results

This analysis explored how to predict the risk of customers missing or delaying payments when using Buy Now Pay Later (BNPL) services. The results show that it is possible to build models that are both useful and reasonably easy to understand.

First, repayment risk can be modeled clearly using approaches like logistic regression. The baseline model achieved an accuracy of about **64%** and a ROC-AUC score of around **0.71**,

meaning it could reasonably distinguish between customers who will default and those who will not. This model provides a simple and interpretable way to understand how different factors affect risk.

```
Baseline Logistic Regression Results
-----------------------------------
Accuracy : 0.6428
ROC-AUC  : 0.7061

Classification Report:
              precision    recall  f1-score   support

           0       0.87      0.63      0.73      7680
           1       0.36      0.68      0.47      2320

    accuracy                           0.64     10000
   macro avg       0.61      0.65      0.60     10000
weighted avg       0.75      0.64      0.67     10000
```

Second, **uncertainty in predictions can be measured using the Bayesian approach.** The Bayesian model achieved a higher overall accuracy of about **78%**, with a similar ROC-AUC score of around **0.71**. This means the overall prediction accuracy improved, while still maintaining a similar ability to separate risky and non-risky customers. Unlike the baseline model, the Bayesian approach also shows how confident the model is in its predictions by considering a range of possible outcomes rather than a single fixed estimate.

```
Bayesian Logistic Regression (FULL POSTERIOR)
--------------------------------------------
Accuracy: 0.7848
ROC-AUC: 0.7058

Classification Report:
              precision    recall  f1-score   support

           0       0.79      0.99      0.88      7680
           1       0.75      0.11      0.19      2320

    accuracy                           0.78     10000
   macro avg       0.77      0.55      0.53     10000
weighted avg       0.78      0.78      0.72     10000
```

Third, several key factors were found to strongly influence repayment risk. Customers with lower credit scores and lower income were much more likely to default, while higher credit scores significantly reduced risk. For example, credit score had the strongest negative impact on default risk (around **-0.65**), making it the most important protective factor. On the other hand, factors like using a VPN connection (around **+0.32 in the Bayesian model**) and certain BNPL providers were associated with higher risk. These results were consistent across both modeling approaches.

	Feature	Logistic_Coefficient	Bayesian_Coefficient
2	Credit_Score	-0.659677	-0.654778
10	Connection_Type_VPN	1.019903	0.324922
1	Annual_Income	-0.074392	-0.080920
8	BNPL_Provider_Afterpay	0.140179	0.064650
7	Purchase_Category_Home & Furniture	-0.093692	-0.036434
9	Device_Type_Mobile	0.062909	0.034818
4	Checkout_Time_Seconds	-0.021377	-0.024460
11	Connection_Type_WiFi	-0.048572	-0.022398
0	Customer_Age	0.021774	0.020984
5	Gender_Male	-0.040684	-0.018644
6	Purchase_Category_Electronics	0.030957	0.008637
3	Purchase_Amount	0.007904	0.006948

Fourth, ethical concerns are important when using automated systems like this. The models showed some weaknesses, especially in identifying high-risk customers. For example, in the Bayesian model, although overall accuracy was high (78%), it only correctly identified a small portion of actual default cases (low recall). This means many risky customers were missed. At the same time, earlier models sometimes flagged too many safe customers as risky. These issues highlight the importance of not relying entirely on automated decisions and ensuring fairness in how predictions are used.

Finally, the project demonstrates how a responsible and reproducible approach to financial modeling can be developed using tools like Python. The process was transparent, structured, and easy to follow, making it possible for others to review and improve the work. This supports better collaboration and trust in the results.

In summary, the models were effective at identifying general patterns in repayment risk, with accuracy ranging from **64% (baseline)** to **78% (Bayesian)**. However, all models struggled to consistently detect high-risk customers. The Bayesian approach added value by improving

stability and showing uncertainty, but further improvements are needed to better capture default cases.

6 Critique of the model

6.1 Does the model satisfactorily answer the questions posed?

Overall, the model satisfactorily answers the questions posed in Section 1.3. It provides a clear and appropriate framework for studying BNPL repayment risk, and it addresses the main project goals in a way that is mathematically sound, interpretable, and reproducible. At the same time, the model should be viewed as a strong first step rather than a complete solution for real-world deployment.

1. This question was answered satisfactorily because the modelling framework is clear, interpretable, and well matched to the problem. The use of logistic regression in both classical and Bayesian form makes the probability of repayment failure easy to define and explain.
2. This question was answered strongly because uncertainty was incorporated directly into the analysis rather than treated as an afterthought. The Bayesian framework added an important layer of information by allowing uncertainty in parameters and predictions to be quantified explicitly.
3. This question was also answered satisfactorily because the model was able to determine which predictors were most relevant to repayment risk. This gave useful insight into the structure of the problem and supported the overall objective of understanding repayment behavior, not only predicting it.
4. This question was addressed in a meaningful way because the modelling choices emphasized interpretability, transparency, and caution in automated decision-making. Although ethical concerns cannot be fully resolved by a single model, the approach taken here supports more responsible use than a black-box method would.
5. This question was answered satisfactorily through the overall workflow used in the project. The combination of Python, documented preprocessing, structured modelling steps, and version-controlled collaboration created a process that is transparent, reproducible, and suitable for further development.

In summary, the model does answer the main questions posed at the beginning of the project. Its main strength is that it balances interpretability, uncertainty quantification, and reproducibility in a coherent framework. While further refinement would improve its practical usefulness, it successfully meets the core objectives of the project.

6.2 Limitations

While the models produced useful results, there are several important limitations to consider.

One major limitation is the difficulty in identifying high-risk customers. Even though the Bayesian model achieved a relatively high accuracy of around 78%, it failed to correctly detect many actual default cases. This means a large number of risky customers were missed. On the other hand, the baseline model (64% accuracy) sometimes incorrectly labeled safe customers as risky. This trade-off shows that the models are not yet reliable for critical financial decisions.

Another key issue is imbalanced data. Only about 23% of the cases were defaults while the majority were non-defaults. Because of this, the models tend to focus more on predicting non-defaults correctly and struggle to identify defaults.

The models are also limited by the features used. They only include variables like income, credit score, and transaction details. Important real-world factors, such as repayment history, employment status or broader economic conditions, were not included which reduces the model's accuracy.

There are also ethical concerns. The models can make incorrect predictions which may unfairly affect customers. For example, some individuals may be wrongly classified as high risk, while others who are actually risky may be overlooked. This highlights the need for careful use and human oversight when applying such models.

6.3 Future Improvements

If more time were available, several improvements could be made to enhance the model. First, techniques to handle imbalanced data, such as resampling, adjusting class weights further, or using cost-sensitive learning, could improve the model's ability to detect default cases.

Second, more advanced machine learning models, such as decision trees, random forests or gradient boosting methods, could be explored. These models may better capture complex patterns in the data and improve prediction performance.

Third, adding more detailed and realistic data would likely improve results. Including variables like past repayment behavior, customer spending trends or external economic indicators could provide a more complete picture of risk.

Fourth, more focus could be placed on fairness and transparency. This includes checking for potential bias in predictions and ensuring that the model's decisions can be clearly explained and justified.

Finally, the Bayesian approach could be extended further by making full use of uncertainty in decision-making, rather than relying only on fixed thresholds. This would allow for more cautious and informed financial decisions.

7 Appendix: Supporting Material

All code, data processing steps, and model implementations used in this project are available in the following GitHub repository:

Repository Link:
https://github.com/samarapires-ml/Buy-Now-Pay-Later--BNPL--Risk-Prediction

Key notebooks included in the repository are:

- Data Cleaning
- Exploratory Data Analysis (EDA)
- Feature Engineering
- Model Development (Please Note: Due to file size constraints, GitHub may not render this notebook and may display an **"invalid notebook"** warning. This does not indicate an error - please download the notebook locally for proper viewing.)